Auteur: Xander De Blauwer
Studierichting: Informatica
Oefeningenreeks 4A nr. 63.5

$$= \int \frac{dx}{(\sqrt{1+x^2})^3}$$

Stel $x = \tan(t) \Rightarrow dx = \sec^2(t)dt$

$$= \int \frac{\sec^2(t)}{\sqrt{\sec^2(t)}^3} dt$$

$$= \int \frac{dt}{\sec(t)} = \int \cos(t)dt$$

$$= \sin(t) + c = \sin(\mathrm{Bgtan}(x)) + c$$

Stel $\theta = \mathrm{Bgtan}(x) \Rightarrow \tan(\theta) = x$

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\Rightarrow x = \frac{\sin(\theta)}{\cos(\theta)} \Leftrightarrow \sin(\theta) = x\cos(\theta)$$

$$x^2\cos^2(\theta) + \cos^2(\theta) = 1$$

$$\Rightarrow \cos(\theta) = \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow \sin(\theta) = \frac{x}{\sqrt{1+x^2}}$$

$$\int \frac{dx}{(\sqrt{1+x^2})^3} = \frac{x}{\sqrt{1+x^2}} + c$$

References